

Indian Institute of Information Technology, Design & Manufacturing Kancheepuram,
Melakottaiyur, Chennai-127.

Answer All Questions. Marks for Individual Questions are indicated in parenthesis

Computer Vision(CS5100)

Mid Sem Exam, Sep 2025

Time: 120 Mins.

Max. Marks: 20

1. Given an algorithm, how will you check whether it is a computer vision algorithm or not? Justify your answer (1)
2. Suppose you have a built-in function for computing correlation, how will you use it to find convolution?. Justify your answer. (1)
3. Does **Sobel filter** filter noise before finding derivative? Justify your answer (2)
4. What is the difference between segmentation and object detection (1)
5. What is the time complexity of computing convolution using brute force method(computing using definition of convolution), assuming the image size is $n \times n$ and filter size is $m \times m$. Justify your answer (1)
6. Give Canny Edge Detection Algorithm, and write the novelty of the algorithm, also write for what kind of images this algorithm will work well with justification (4)
7. Justify why weighted centroid is preferred over centroid in Mean Shift Segmentation Algorithm(MSSA), also derive the optimum weights that are used in weighted centroid in MSSA (3)
8. For what kind of images, each of the following segmentation algorithm will work well. Justify your answer
 - Thresholding based segmentation
 - Mean Shift Segmentation
 - Watershed Segmentation
 - Graph-Cut Segmentation(2)
9. Given an image how will you construct input for Graph-cut Segmentation Algorithm (2)
10. Given two parallel cameras(Image planes are coplanar), and the right camera is translated by 10 cm from the left camera along X -direction. Find the **essential matrix**, and also find the **equation of the line** on the image plane of the right camera for the point (2, 5) in the image plane of the left camera such that the corresponding point for (2, 5) lies on the line (3)